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Provisional Patent Application

for:

MULTIFUNCTIONAL CHITOSAN HYDROGEL WOUND DRESSING FOR SUSTAINED ANTIMICROBIAL AND VASODILATORY THERAPY IN SICKLE CELL ULCERS

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MULTIFUNCTIONAL CHITOSAN HYDROGEL WOUND DRESSING FOR SUSTAINED ANTIMICROBIAL AND VASODILATORY THERAPY IN SICKLE CELL ULCERS

TECHNICAL FIELD

The present invention relates to compositions useful for treating pathologies such as wounds occurring in sickle cell anemia patients, and methods for making and using these compositions.

BACKGROUND OF THE INVENTION

Chronic non-healing wounds affect approximately 1–2% of the world's population[1]. These wounds are classified as chronic based on their inability to heal within an expected time frame1. They typically occur due to complications from other diseases such as sickle cell anemia, diabetes, neuropathies, and arterial insufficiency. Other factors that influence their formation include burns, trauma, and surgical complications[1]. Among chronic wounds, sickle cell leg ulcers (SCLUs) are particularly challenging to treat due to their complexity and unique pathophysiology. Their pathophysiology stems from the underlying driver of sickle cell disease (SCD) – vaso-occlusion; where vascular blockages, low nitric oxide (NO) bioavailability and hemolysis trigger and prolong ulceration. SCLU manifestation evolves from random pain to subcutaneous necrosis, blister formation, and then epidermis destruction which is known to persist for up to 20 years[2,3]. They are also found on the lower extremities of the body – regions already predisposed to poor blood circulation, which exacerbates delayed healing and increases the risk of chronicity[1]. Furthermore, these wounds are often heavily colonized by skin microflora such as Staphylococcus aureus and Pseudomonas aeruginosa, which become opportunistic pathogens and contribute to their longevity.

The standard for SCLU treatment largely involves therapies that primarily target bacterial colonization while failing to address the underlying pathophysiology. These include hydrocolloids, silver- and iodine-based dressings, and standard protective bandages4. While infection control is critical, addressing only this component overlooks vaso-occlusion that leads to SCLU persistence. Recent work by Caterina Minniti has aimed to address this through the use of topical sodium nitrite as a vasodilator and NO donor, to improve blood flow in SCLUs and target vaso-occlusion[5,6]. However, topical applications face significant limitations,

including poor control with dosage, the need for frequent reapplication, and a lack of sustained release.

The above-noted limitations in conventional therapies may be overcome by an integrated therapy – one that provides sustained antimicrobial and vasodilatory effects while remaining biocompatible with the mechanically active skin of the lower extremities.